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Information Foraging in the Attention Economy and the Rising Entropy of English - Supplementary Information

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1 Supplementary Information — Linguistic Niche Hypothesis

The finding in the main paper of word entropy, and lexical diversity, rising in American English is the opposite of what might be predicted by the Linguistic Niche Hypothesis. That hypothesis makes predictions about the complexity of language morphology (e.g. I ate, la casita) and syntax (e.g. I did eat, la pequeña casa), with the assumption that complexity is balanced between the two. The Linguistic Niche Hypothesis [**?**] suggests that languages in large, spread out social systems tend to have simpler morphological forms, with the grammatical work instead being done through syntax [**?**]. The hypothesised mechanism for this is that second language learners prefer simpler forms so that complex morphological forms disappear over time [**?**]. A global lingua franca like English should therefore be undergoing morphological simplification, and evidence does suggest that this is the case with the regularisation of English past tense verbs [**?**, **?**] and a loss of inflectional diversity [**?**]. Further work suggests that this morphological simplification should correlate with a reduction in lexical diversity as measured by type token ratio [**?**, **?**] (or word entropy) — complex morphological forms are non-repetitive (many unique word types per word token) whilst syntactic grammatical modifiers are repetitive (few unique word types per word token). We find that lexical diversity is instead rising in American English. We suggest some possible explanations:

1. English morphology is overall becoming more complex, against the Linguistic Niche Hypothesis.
2. English morphology is becoming simpler without an increase in syntactic complexity. This would be a further refutation of the already beleaguered [**?**, **?**] equicomplexity assumption, which states that mature languages have broadly equal grammatical complexity, balanced between morphology and syntax.
3. Lexical diversity (and Type Token Ratio) is not a good measure of morphological complexity. The increase in lexical diversity is instead driven by more concise information and a wider, and faster switching of, contexts in written media.

The third option here aligns well with the ideas in the main paper, and is in our opinion at least partly responsible. If people are drawn towards higher utility rate information then that could drive English to be more concise and to switch contexts more quickly.

2 Supplementary Information - Historical Analysis of US Magazine Publishing

As a case study we investigated the history of magazine publishing in America. Figure 1 shows the historical trend in COHA magazine word entropy alongside magazine circulation figures and important

events. Magazine publishers are in a two-sided market where they sell magazines to consumers and attention to advertisers [**?**], with the majority of revenue from selling attention [**?**]. This wasn't always the case in the US — prior to the 1890s most magazine revenue was from sales, with advertising considered undesirable [**?**]. Towards the late 19th century a combination of rapidly decreasing printing costs, growth in the literate population, discounts from the US postal service and the ability to target adverts to a niche readership led to a new business model to emerge [**?**]. This new model involved selling magazines lower than the price of production, which increased circulation so that those costs could be recouped by advertising revenue [**?**]. Before 1893, most magazines sold for 25 cents — until a price war led to the magazines McClure's, Munsey's and Cosmopolitan dropping their prices to 10 cents and subsequently enjoying rises in circulation and advertising revenue [**?**]. The 10 cent magazines contributed to a tripling in total magazine readership from 1890 in 1905 [**?**], and there was a huge jump in word entropy in the same period (Figure 1).

The Audit Bureau of Circulation was created by advertisers in 1914 [**?**] to more accurately measure magazine readership numbers. This quantification of attention further increased pressure on magazine publishers to improve their circulation numbers in order to sell advertising. Other changes included moving advertisements from the back of the magazine to alongside the main content — a move that forced copywriters to improve the appeal of the content through adding color and improving graphics [**?**].

Word entropy continues to rise throughout the 20th century alongside magazine circulation, with a Pearson's correlation coefficient r= 0.91 ($p < 0.001$), although both rise over time so that confounding factors are not ruled out (Figure 1). After the 1890s, the biggest drop in word entropy was during the great depression when magazine circulation also fell. There is a suggestion in the data that things change around the year 2000, as magazine circulation drops but word entropy continues to rise. The rise of digital media around this time is perhaps the biggest change in publishing since the printing press so we would not expect the same trends to necessarily continue — and digital media represents a new competitive pressure.

3 Supplementary Information — Prey Choice Model Derivation

In the main paper we justify the prey choice algorithm using an argument that considers the opportunity cost of spending time handling a prey versus searching in the environment. Here we derive the same result more rigorously. This is a completely analogous derivation as found in optimal foraging theory [**?**]. As in the main paper, we have information types, i, that are encountered with rates λ_i while searching. Each information item, if consumed, provides a benefit u_i in a handling time t_i, during which the forager is not searching for other items.

In the main text, a media patch expected utility rate is given by,

$$R_{media} = \frac{\sum_D \lambda_i u_i}{1 + \sum_D \lambda_i t_i} \,. \tag{1}$$

This assumes that information types are either in the diet, D, in which case they are always consumed upon encounter, or alternatively the items are not in the diet and never consumed. We can generalise this so that forager's have some probability of consuming an information type upon encounter, p_i,

$$R_{media} = \frac{\sum \lambda_i u_i p_i}{1 + \sum \lambda_i t_i p_i} \,. \tag{2}$$

The forager can choose the probability of paying attention to each information type, and a forager's strategy can be defined as a vector $\mathbf{p} = [p_1, p_2, ..., p_n]$. These choices are independent. To find the strategy that gives the maximum utility rate we can consider each of these choices, p_j, independently. To find the best strategy we separate p_j from the summations and differentiate

$$\frac{\partial R_{media}}{\partial p_j} = \frac{\lambda_j u_j (1 + p_j \lambda_j t_j + \sum_{i \neq j} p_i \lambda_i t_i) - \lambda_j t_j (p_j \lambda_j u_j + \sum_{i \neq j} p_i \lambda_i u_i)}{(1 + p_j \lambda_j t_j + \sum_{i \neq j} p_i \lambda_i t_i)^2} \,. \tag{3}$$

Cancelling like terms

$$\frac{\partial R_{media}}{\partial p_j} = \frac{\lambda_j u_j (1 + \sum_{i \neq j} p_i \lambda_i t_i) - \lambda_j t_j (\sum_{i \neq j} p_i \lambda_i u_i)}{(1 + p_j \lambda_j t_j + \sum_{i \neq j} p_i \lambda_i t_i)^2} \,. \tag{4}$$

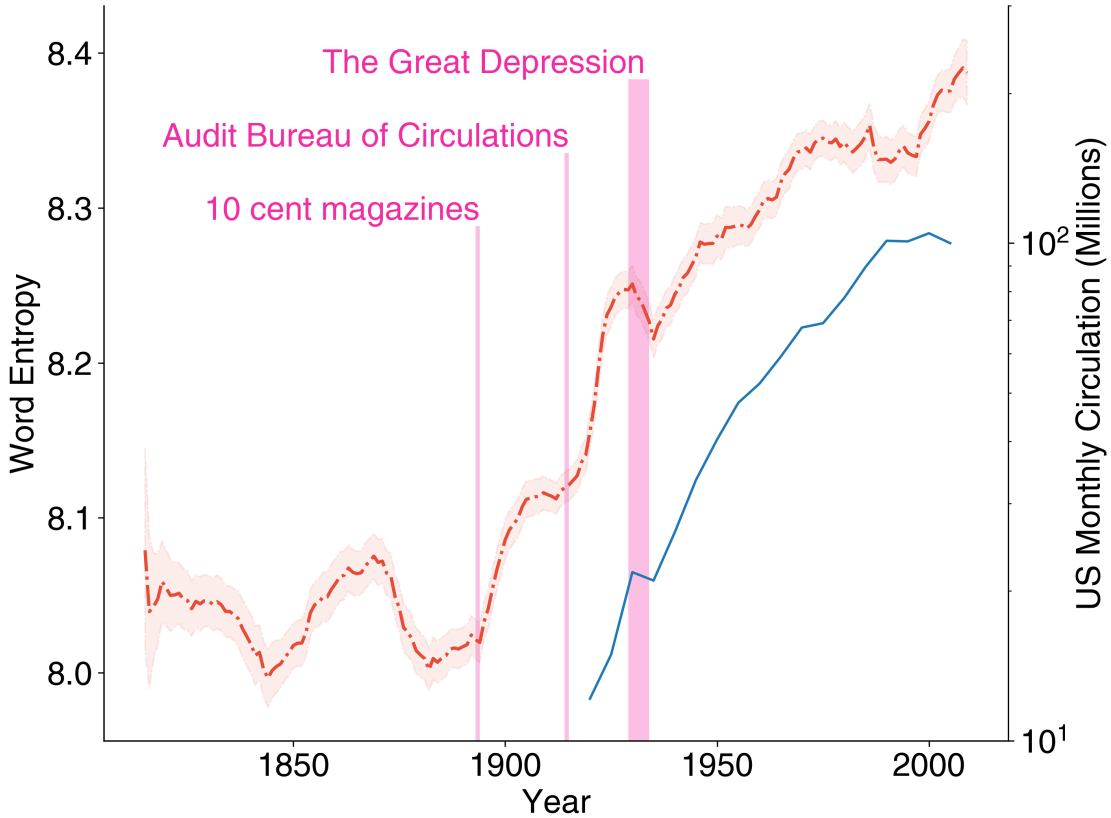

Figure 1: Historical analysis of word entropy in magazines (red dotted, timeseries calculated as in previous figure) with key events (pink) and US Monthly Magazine circulation as reported by the Audit Bureau of Circulations (purple).

The sign of this does not depend on p_j. So if $\frac{\partial R}{\partial p_j} > 0$, R_{media} will be maximised with $p_j = 1$, and otherwise with $p_j = 0$. The condition for $p_j = 1$ is

$$\frac{u_j}{t_j} > \frac{\sum_{i \neq j} p_i \lambda_i u_i}{1 + \sum_{i \neq j} p_i \lambda_i t_i)} \,. \tag{5}$$

The right hand side is the total expected rate of utility for all items except for item j, $R_{\neg j}$. The item should be included in the diet if the utility rate of the item, $r_i = \frac{u_j}{t_j}$, is greater than the overall rate of foraging without the item.

$$r_j \geq R_{\neg j} \,. \tag{6}$$

This is equivalent to the diet inclusion criteria given in the main paper. To find the optimal diet, one can add items in order of their utility rate until the inequality fails.

4 Supplementary Information — Patch Choice Model and Non Constant Patches

The patch choice model considered in the main paper is analogous to the information choice model. Patches of each type are randomly encountered in the environment and encountered as a Poisson processes with rates λ_{media}. We also assume that patches have a constant expected rate of utility, R_{media}, and some finite time, T_{media} until the rate drops to zero, which gives each patch a total utility, U_{media}. Foragers can choose to either consume or ignore a patch upon encountering it. This model is identical to the information choice model so that we can follow that derivation and jump to the conclusion that a